

ECS 5102

Assignment-1



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Started on Wednesday, 17 September 2025, 7:06 PM**State** Finished**Completed on** Wednesday, 17 September 2025, 7:11 PM**Time taken** 4 mins 51 secs**Question 1**

Complete

Marked out of 1.00

Flag question

Which of the following is NOT one of the five classic components of a computer system?

- ☒ a. Compiler
- ☐ b. Input
- ☐ c. Datapath
- ☐ d. Output

Question 2

Complete

Marked out of 1.00

Flag question

The Instruction Set Architecture (ISA) acts as an interface between:

- ☐ a. Application programs and compiler
- ☒ b. Hardware and low-level software
- ☐ c. Operating system and users
- ☐ d. CPU and I/O devices

Question 3

Complete

Marked out of 1.00

Flag question

In a computer system, the bootstrap program is usually stored in:

- ☐ a. Cache
- ☐ b. Hard disk
- ☐ c. Main memory
- ☒ d. ROM/EPROM

Question 4

Complete

Marked out of 1.00

Flag question

Which of the following correctly defines an Operating System (OS)?

- ☐ a. A compiler that converts source code into machine code
- ☐ b. A hardware device managing I/O operations
- ☒ c. A resource allocator and control program for computer hardware
- ☐ d. A program only responsible for word processing

Question 5

Complete

Marked out of 1.00

Flag question

Who introduced binary variables and the fundamental logic operations AND, OR, and NOT?

- ☐ a. John von Neumann
- ☒ b. George Boole
- ☐ c. Alan Turing
- ☐ d. Charles Babbage

Question 6

Complete

Marked out of 1.00

Flag question

What is the decimal equivalent of the binary number 1001_2 ?

- ☐ a. 7
- ☒ b. 9

- ☐ c. 8
- ☐ d. 10

Question 7

Complete

Marked out of 1.00

🚩 Flag question

Which of the following is a single-input logic gate?

- ☐ a. XOR
- ☐ b. OR
- ☒ c. NOT
- ☐ d. AND

Question 8

Complete

Marked out of 1.00

🚩 Flag question

Which of the following instruction formats uses a 26-bit address operand?

- ☒ a. J-Type
- ☐ b. I-Type
- ☐ c. R-Type
- ☐ d. S-Type

Question 9

Complete

Marked out of 1.00

🚩 Flag question

In an R-Type instruction, the funct field is used for:

- ☐ a. Indicating memory addresses
- ☐ b. Storing immediate values
- ☒ c. Identifying the exact operation when opcode is 0
- ☐ d. Specifying the register shift amount

Question 10

Complete

Marked out of 1.00

🚩 Flag question

Which logical instruction is useful for masking bits?

- ☐ a. XOR
- ☒ b. AND
- ☐ c. OR
- ☐ d. NOR

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